

International Economic Development

ECON 4660/8666

Lecture 1: Growth and Development

Who am I?



Google

ZHIGANG FENG - Google Sites

Quantitative Macro · Public Finance · Health & Labor Economics · Machine Learning.

Affiliation

University of Nebraska at Omaha

Research interests

Quantitative Macroeconomics... Labor Economics,...



University of Nebraska

Dr. Zhigang Feng |
College of Business
Administration



A macroeconomist studies AI, public policy and economic theory;



Taught at U of Illinois, U of Zurich, Purdue;



Conduct researches at Congressional Budget Office, Federal Reserve Bank;



Published at top economics journals;



<https://sites.google.com/site/zfeng202/>

2025 Omaha Macroeconomic Forum

About the Forum

The College of Business Administration is proud to launch the 2025 Omaha Macroeconomic Forum, a new premier conference designed to bring together the brightest minds in macroeconomics from across the region. Thanks to a generous donation from Dr. Mark Wohar, UNO/CBA Distinguished Professor, Emeritus, and the crucial support of the CBA Dean's Office, the Forum will showcase frontier research and its practical applications to today's most pressing economic issues. This event is open to the public and offers invaluable insights with direct implications for local business leaders and industry professionals.

Conference Schedule

Date: October 3, 2025

Location: Mammel Hall MH 162, CBA University of Nebraska at Omaha

Address: 6708 Pine Street, Omaha, NE 68182

Thursday, October 2, 2025

6:00 PM Conference Dinner, by invitation

Friday, October 3, 2025

08:00–16:50 Presentation and Discussion

Zoom link and registration: [2025 OMF](https://tinyurl.com/2025OMF)

<https://tinyurl.com/2025OMF>













Course Presentation

- Course website: canvas.
- The choice of textbook.
- Homework, exams and grades.
- Office hours, course materials and additional readings.
- Presentations

Course Requirements

- HomeWorks: Sep 23 – Nov 25, 21%
- Midterm: Monday, Oct 13 (tentative), 18%
- Student-led discussion: Nov 11 – Nov 18, 18%
- Term paper presentation: Nov 25 – Dec 9, 8%
- Attendance: Nov 11 – Dec 9, 18%
- Final: research paper, due Dec. 16, 17%.
- BOOK: [link](#)
- Presentation: [link](#)

Presentation

Delivery

Appropriate volume and rate

Speech is varied to show emphasis and interest

Appropriate posture, eye contact, facial expressions and gestures

Language choice directly and clearly presents topic

Content

Clear and consistent focus with appropriate transitions between main ideas

Speaker demonstrates substantial knowledge of subject

Main ideas are clearly supported through research

Speaker effectively communicates content to audience

Organization

Organized with appropriate beginning, body, and conclusion

Speech organization is clear and logically sequenced

Exhibits research by use of various supporting materials

Message is clearly presented

Evidence of Student Learning

Able to respond intelligently and accurately to questioning

Demonstrates thorough understanding of topic

Clearly explains process used to develop product

Able to reflect on process and communicate learning experience

Line

Bar

Map

Share

Details

Thousand

☒ LABEL

14

12

10

8

6

4

2

0

1960

1970

1980

1990

2000

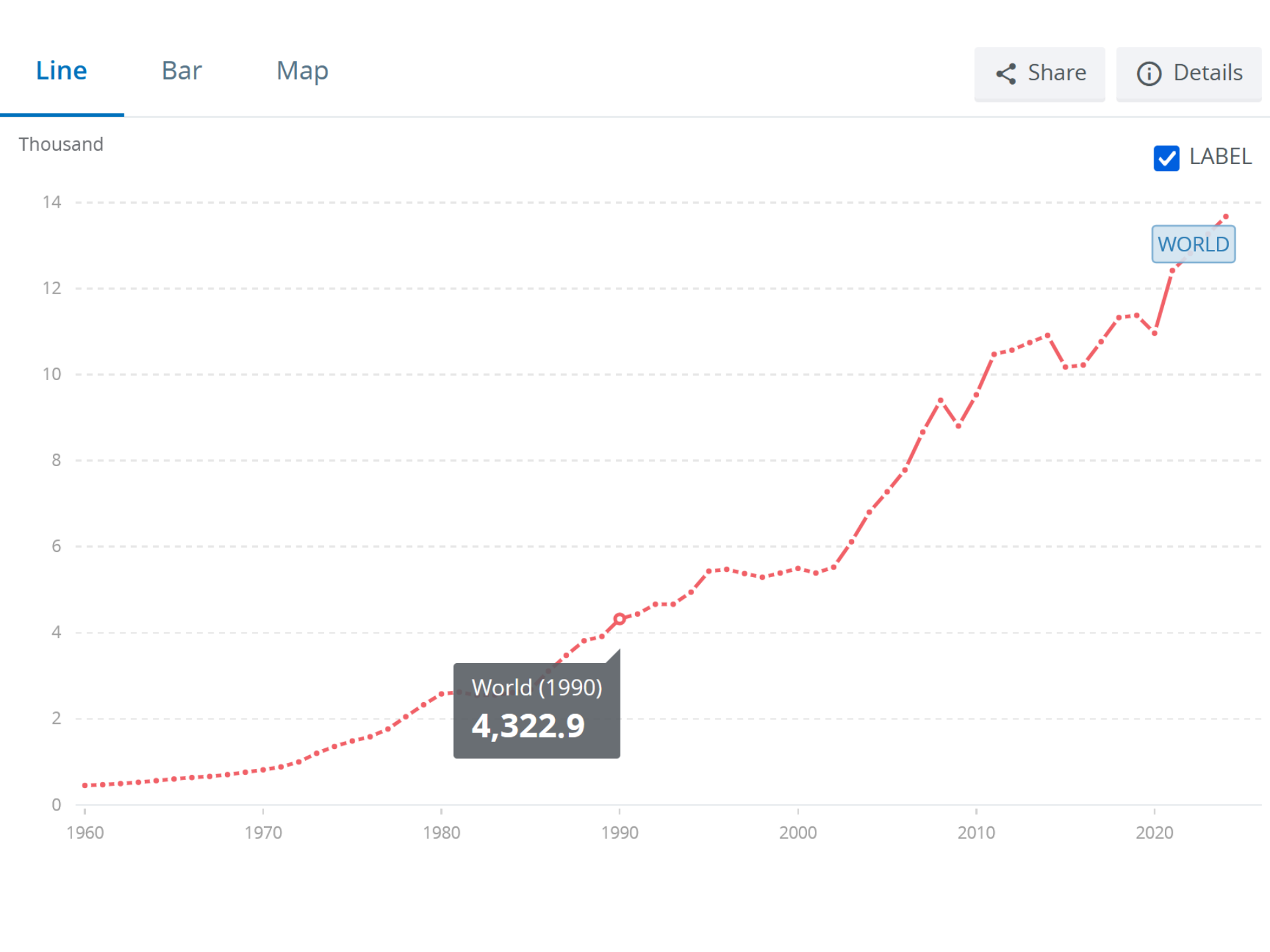
2010

2020

World (1990)

4,322.9

WORLD



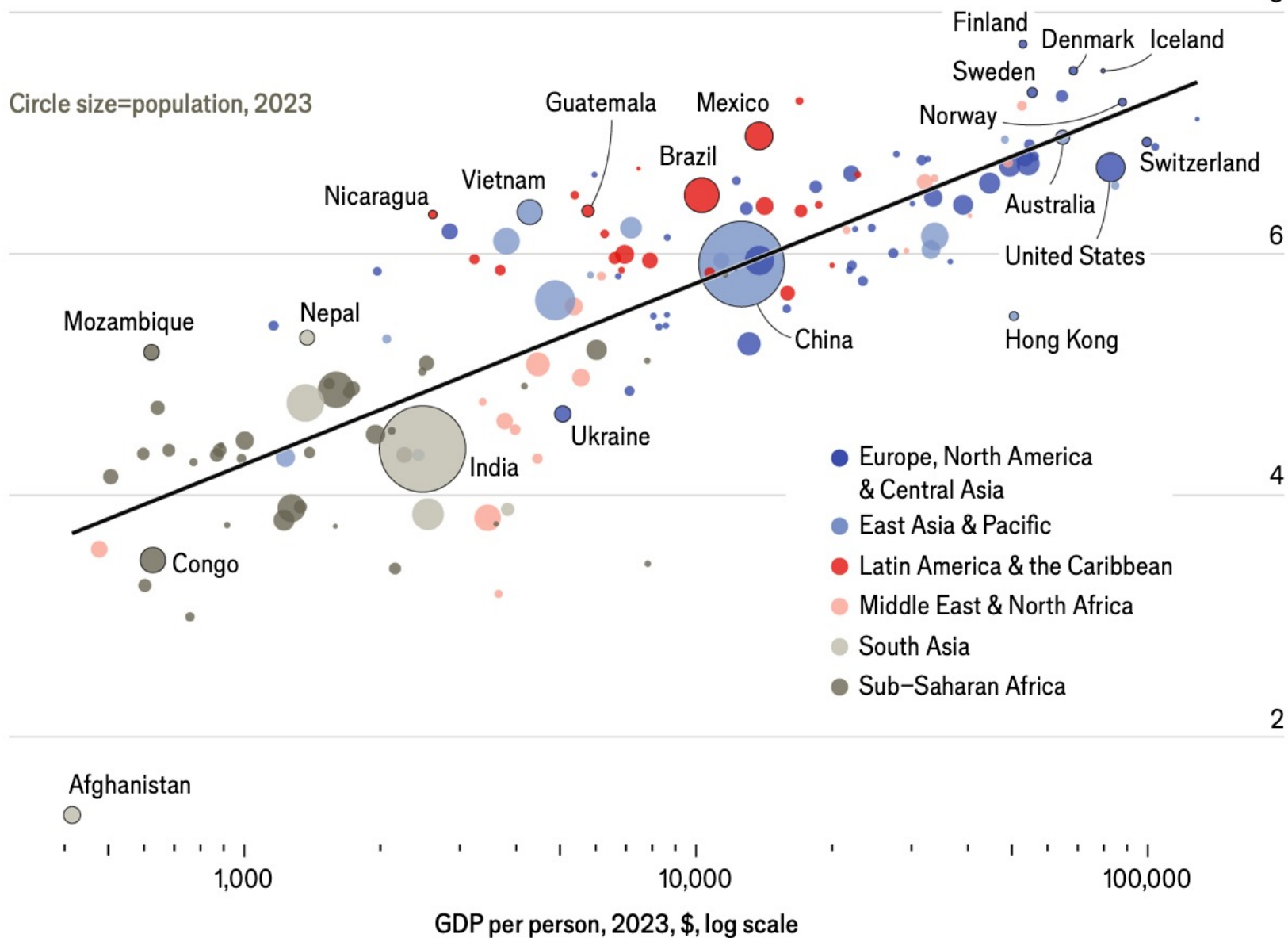
Happiness v GDP per person

Circle size=population, 2023

Life evaluation, 2025

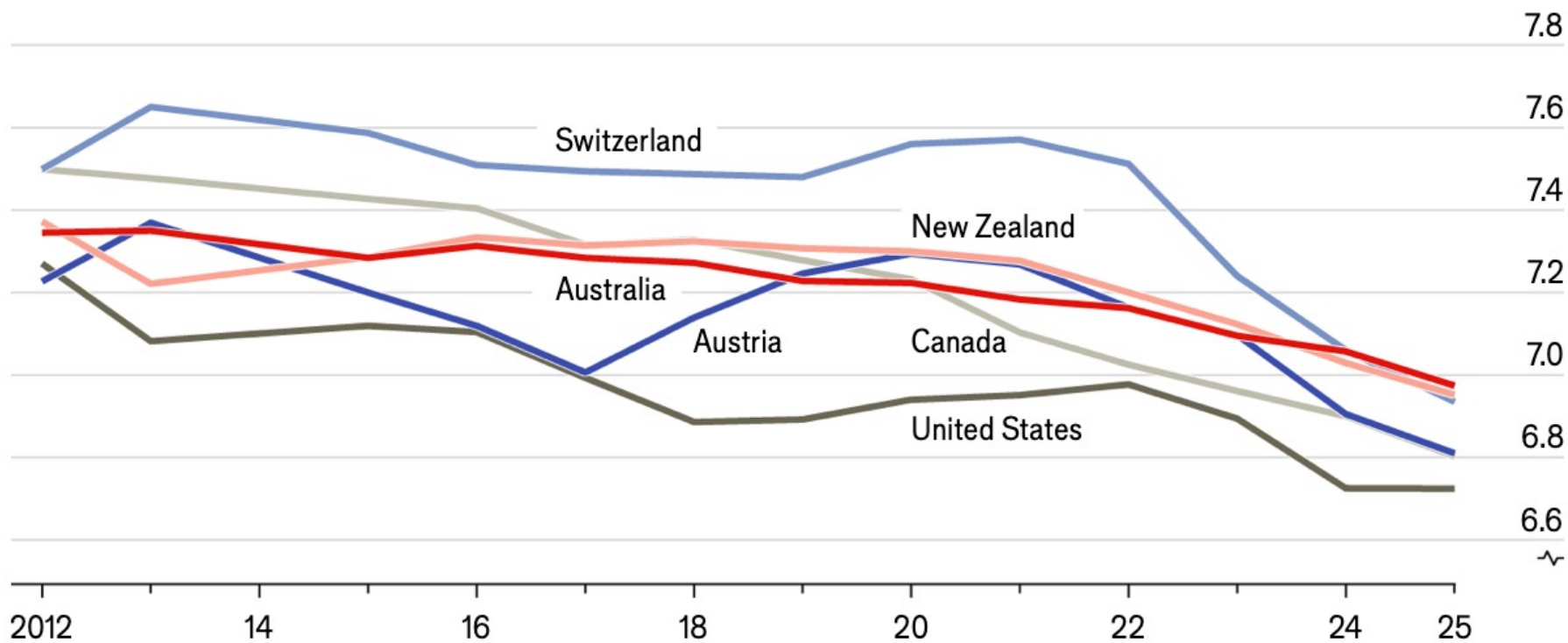
10=happiest

8



Life evaluation, 10=happiest

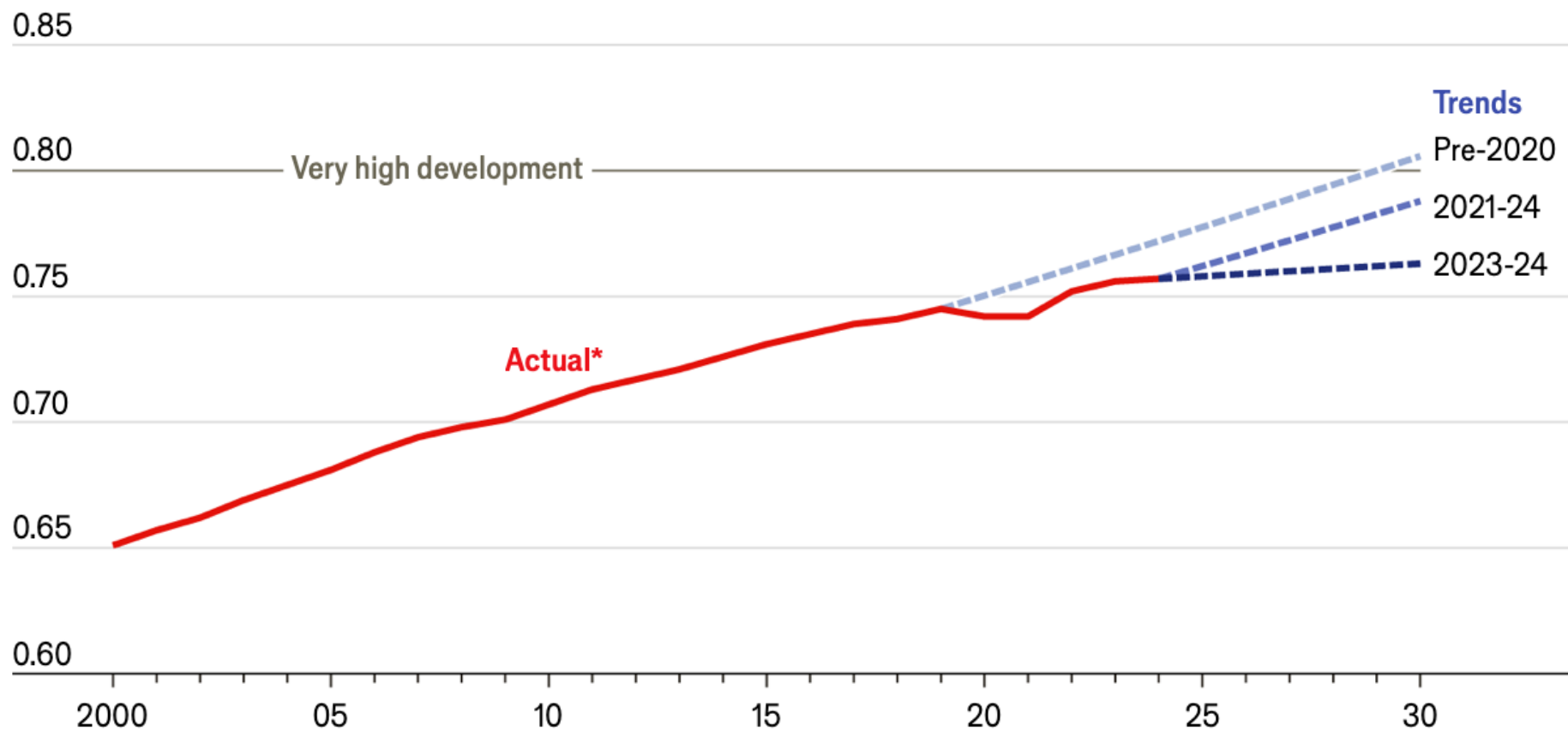
OECD countries with the greatest decline since 2012



Source: World Happiness Report, 2025

Human Development Index, global average

1=most developed



Source: United Nations

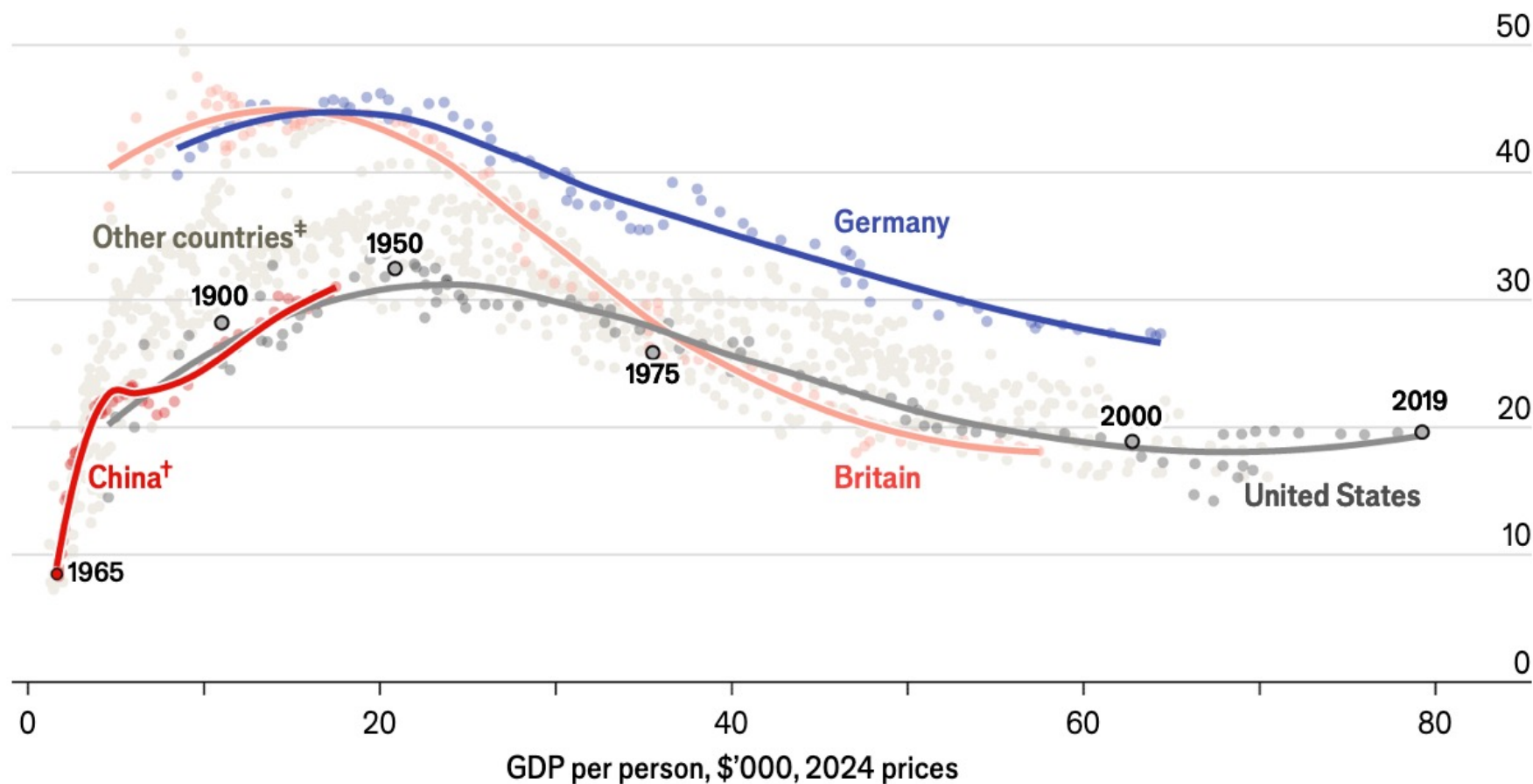
*Estimate for 2024

Human Development Index, 2023

[< Previous](#)[Next >](#)

		Index score, 1=most developed	Life expectancy at birth, years	Expected years of schooling
^		^	◇	◇
1	Iceland	0.972	82.7	18.9
2=	Norway	0.970	83.3	18.8
2=	Switzerland	0.970	84.0	16.7
4	Denmark	0.962	81.9	18.7
5=	Germany	0.959	81.4	17.3
5=	Sweden	0.959	83.3	19.0
7	Australia	0.958	83.9	20.7
8=	Hong Kong	0.955	85.5	16.9
8=	Netherlands	0.955	82.2	18.6
10	Belgium	0.951	82.1	19.0

Industry as % of total employment, 1801-2019*



*1801 or earliest available. 2019 or latest available [†]Post-1965 only

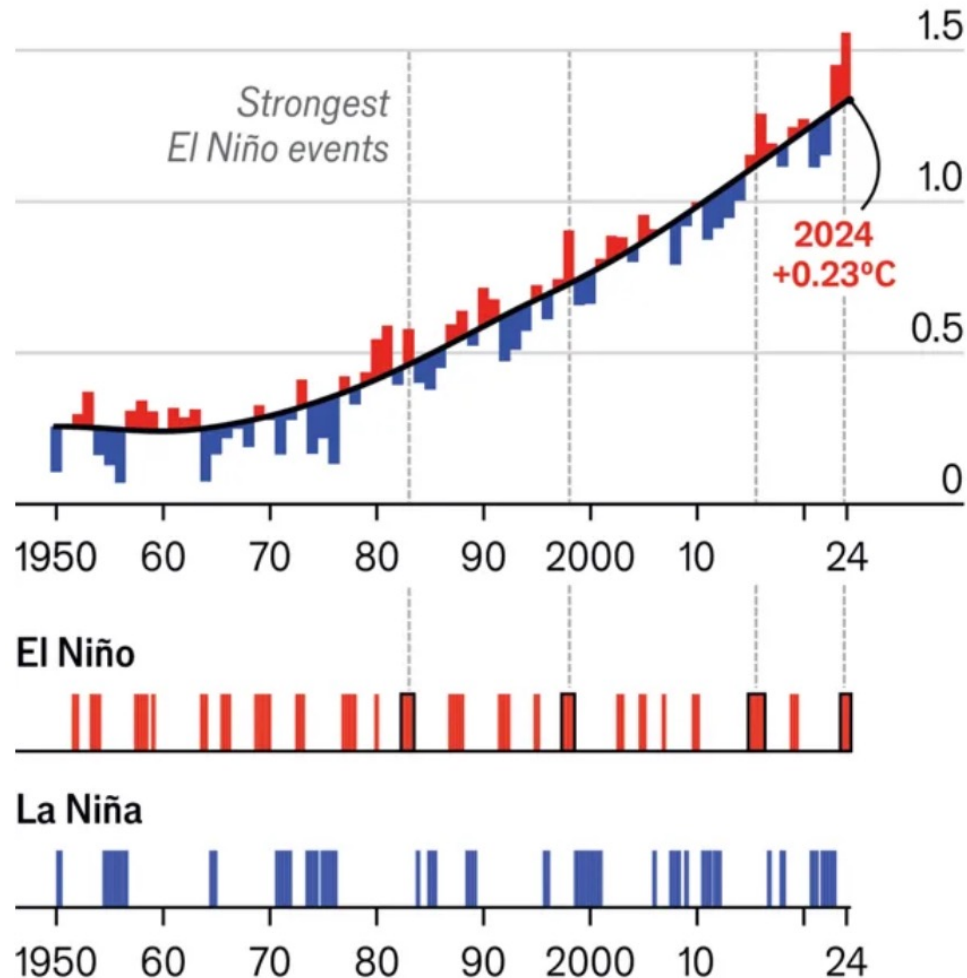
[‡]Belgium, Denmark, Finland, France, Italy, Japan, Netherlands, South Korea, Spain and Sweden

Sources: Groningen Growth and Development Centre; Our World in Data; UN; *The Economist*

Annual deviation from global surface-temperature trend

— Smoothed trendline

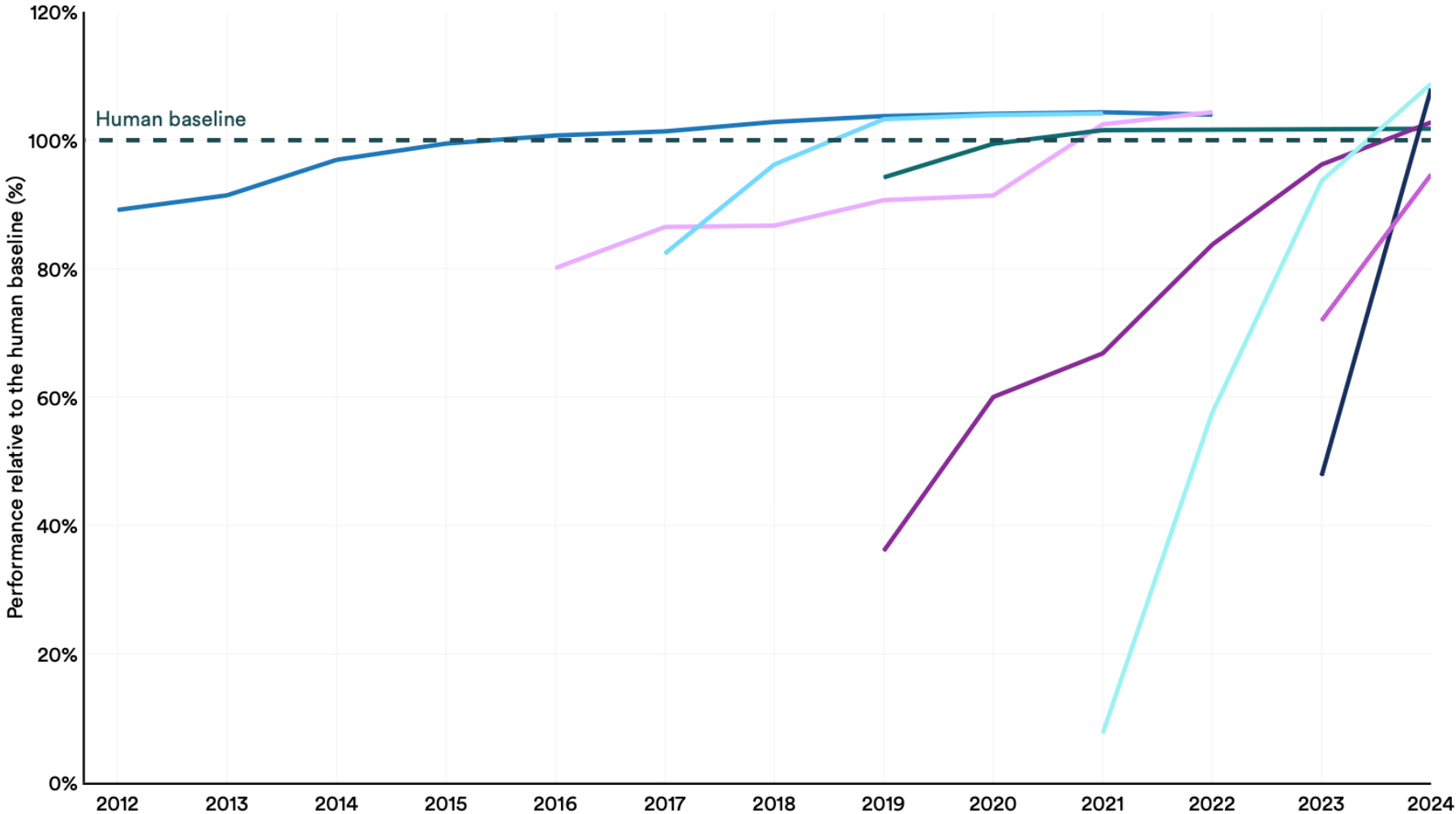
°C from 1850-1900
pre-industrial average



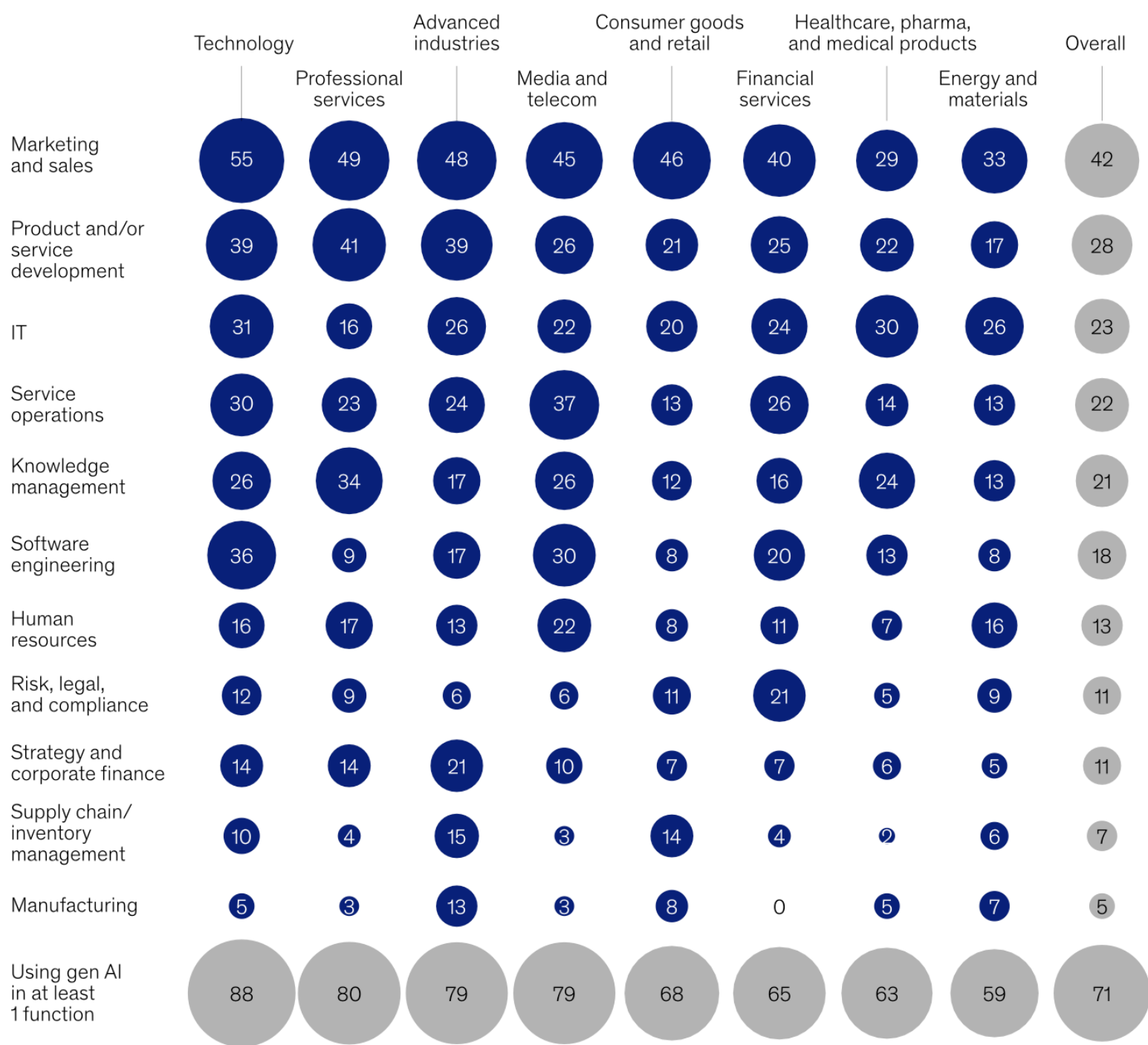
Source: Copernicus, ERA5

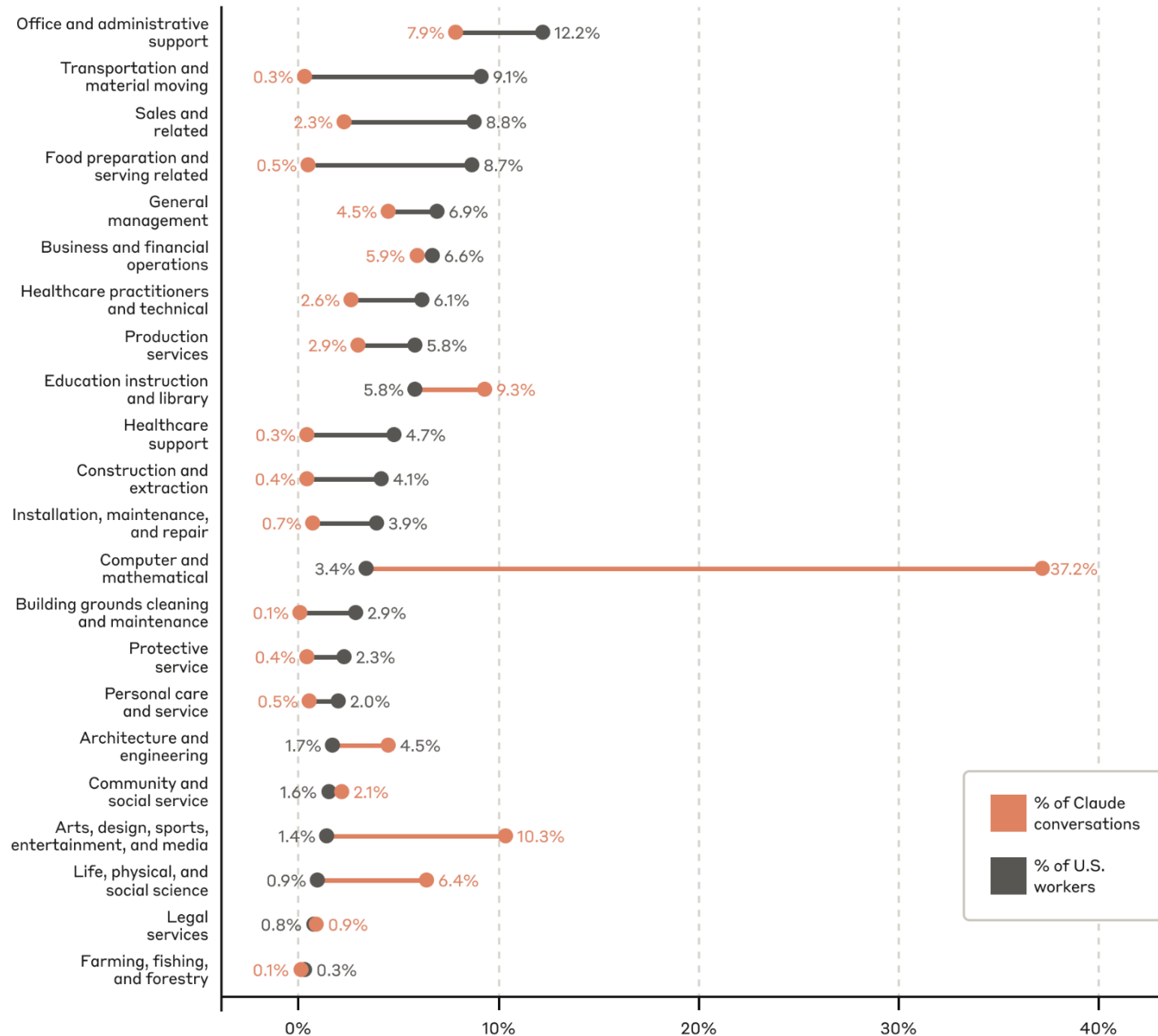
Select AI Index technical performance benchmarks vs. human performance

Source: AI Index, 2025 | Chart: 2025 AI Index report

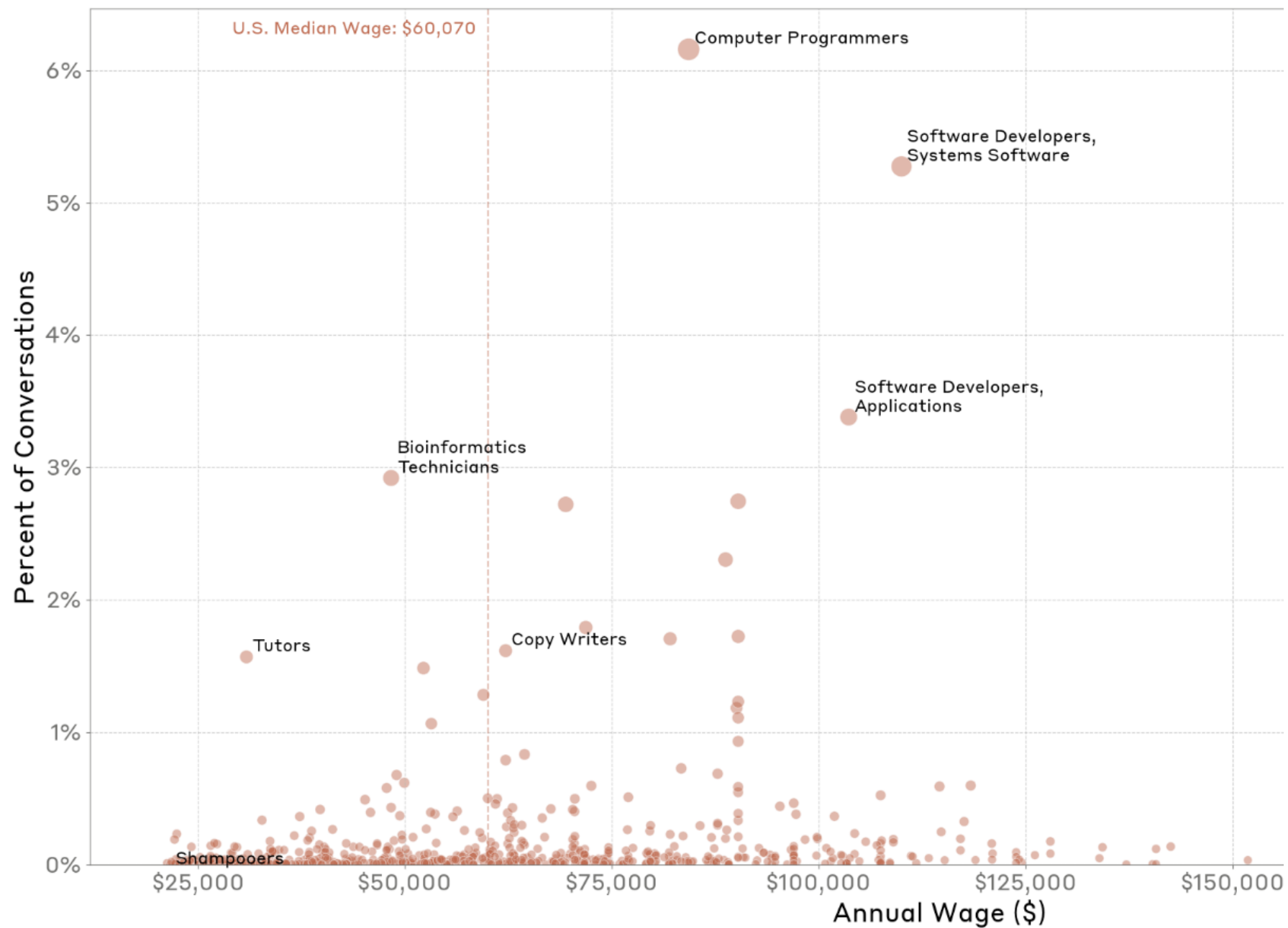


- Image classification (ImageNet Top-5)
- Medium-level reading comprehension (SQuAD 2.0)
- Multitask language understanding (MMLU)
- PhD-level science questions (GPQA Diamond)
- Visual reasoning (VQA)
- English language understanding (SuperGLUE)
- Competition-level mathematics (MATH)
- Multimodal understanding and reasoning (MMMU)



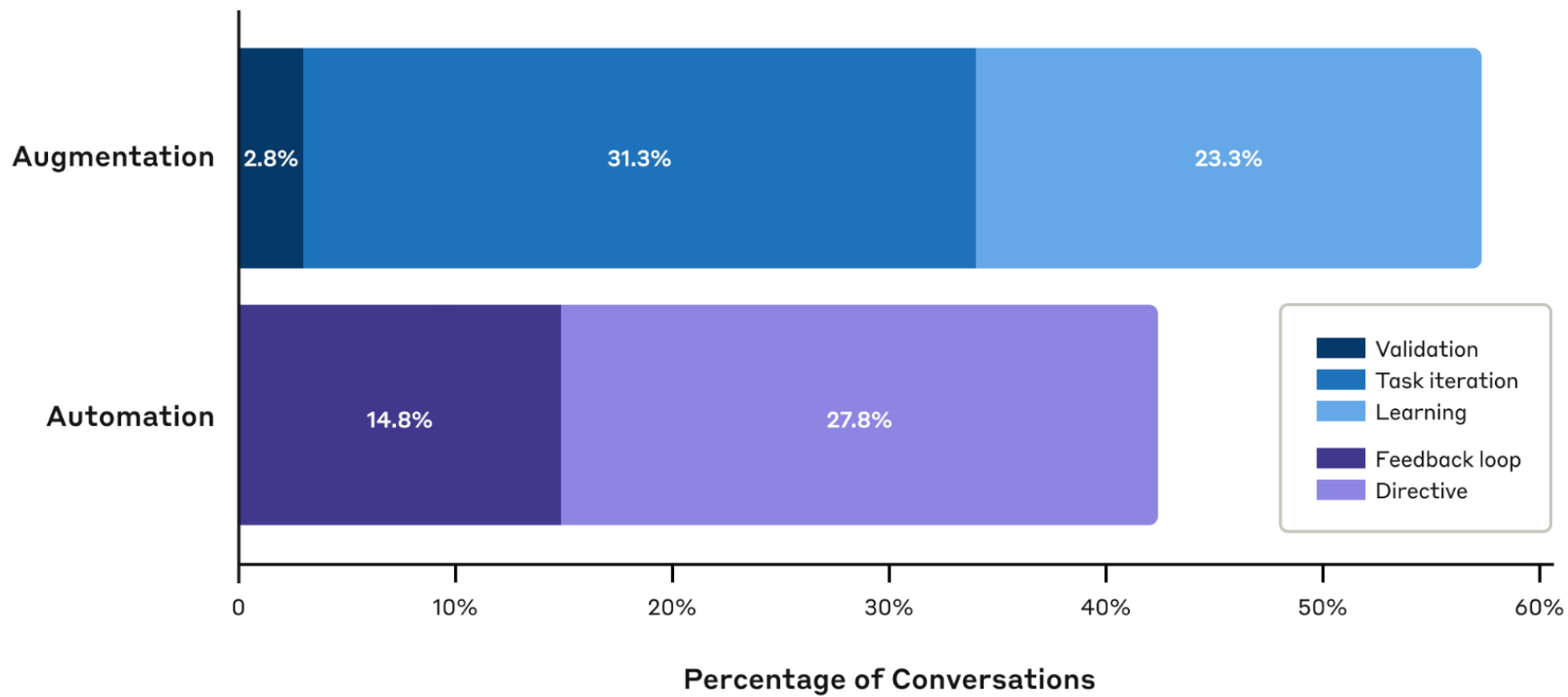


Representation relative to US economy

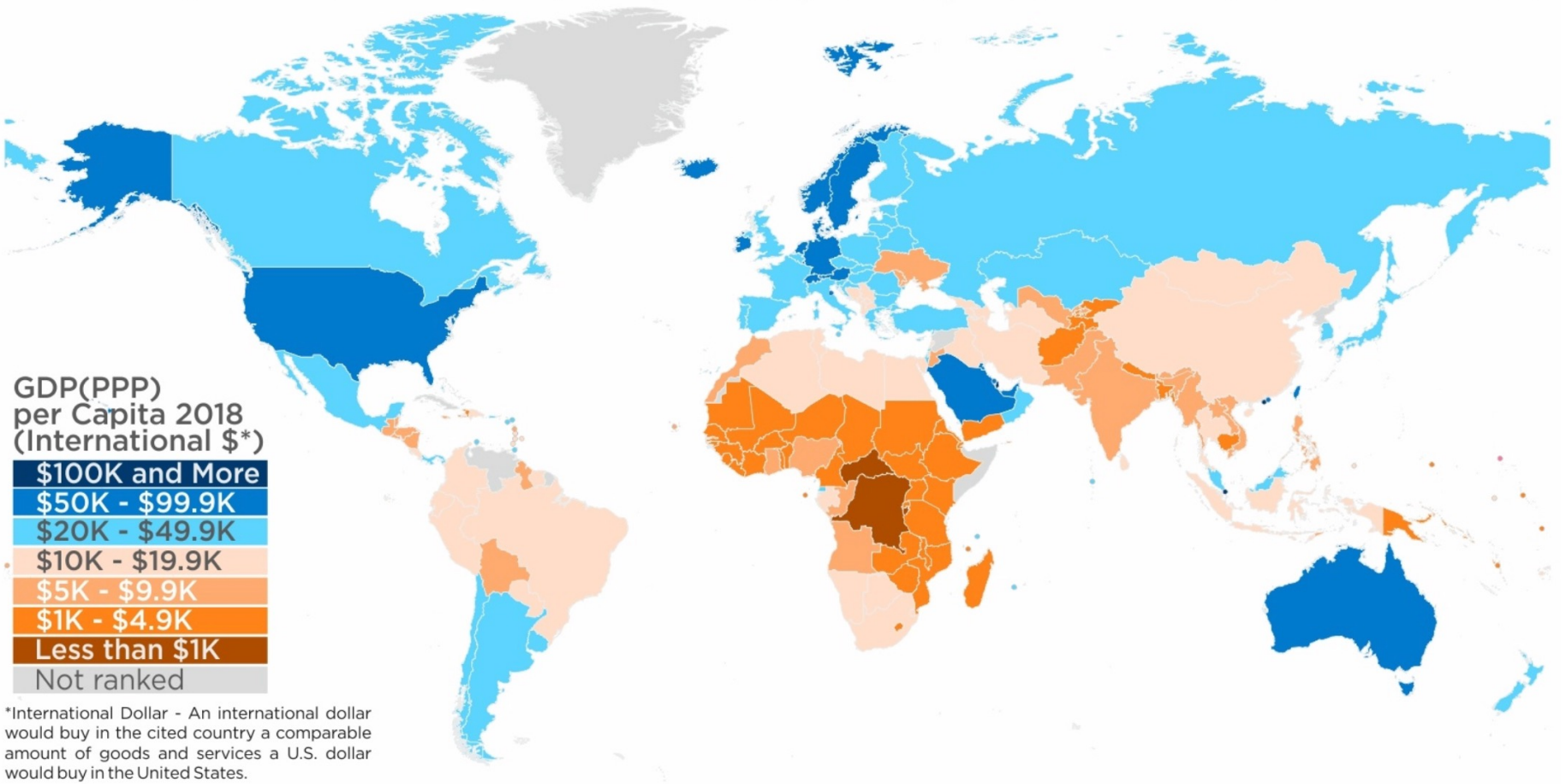


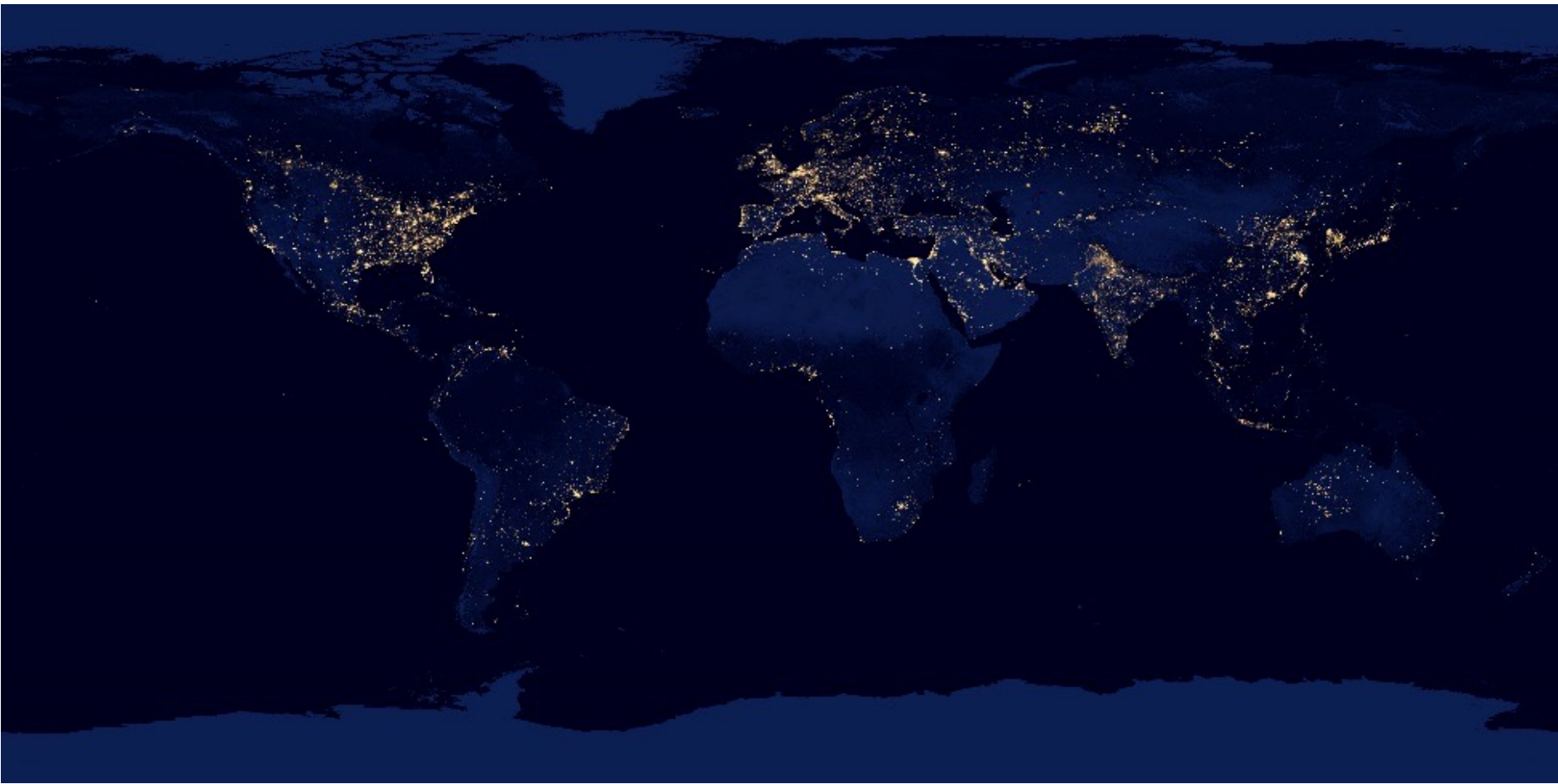
Automative Behaviors	Augmentative Behaviors
<i>AI directly executes tasks with minimal human involvement</i>	<i>AI enhances human capabilities through collaboration</i>
Directive: Complete task delegation with minimal interaction <i>Illustrative Example: “Format this technical documentation in Markdown”</i>	Task Iteration: Collaborative refinement process <i>Illustrative Example: “Let’s draft a marketing strategy for our new product. ... Good start, but can we add some concrete metrics?”</i>
Feedback Loop: Task completion guided by environmental feedback <i>Illustrative Example: “Here’s my Python script for data analysis – it’s giving an IndexError. Can you help fix it? ... Now I’m getting a different error...”</i>	Learning: Knowledge acquisition and understanding <i>Illustrative Example: “Can you explain how neural networks work?”</i>
	Validation: Work verification and improvement <i>Illustrative Example: “I’ve written this SQL query to find duplicate customer records. Can you check if my logic is correct and suggest any improvements?”</i>

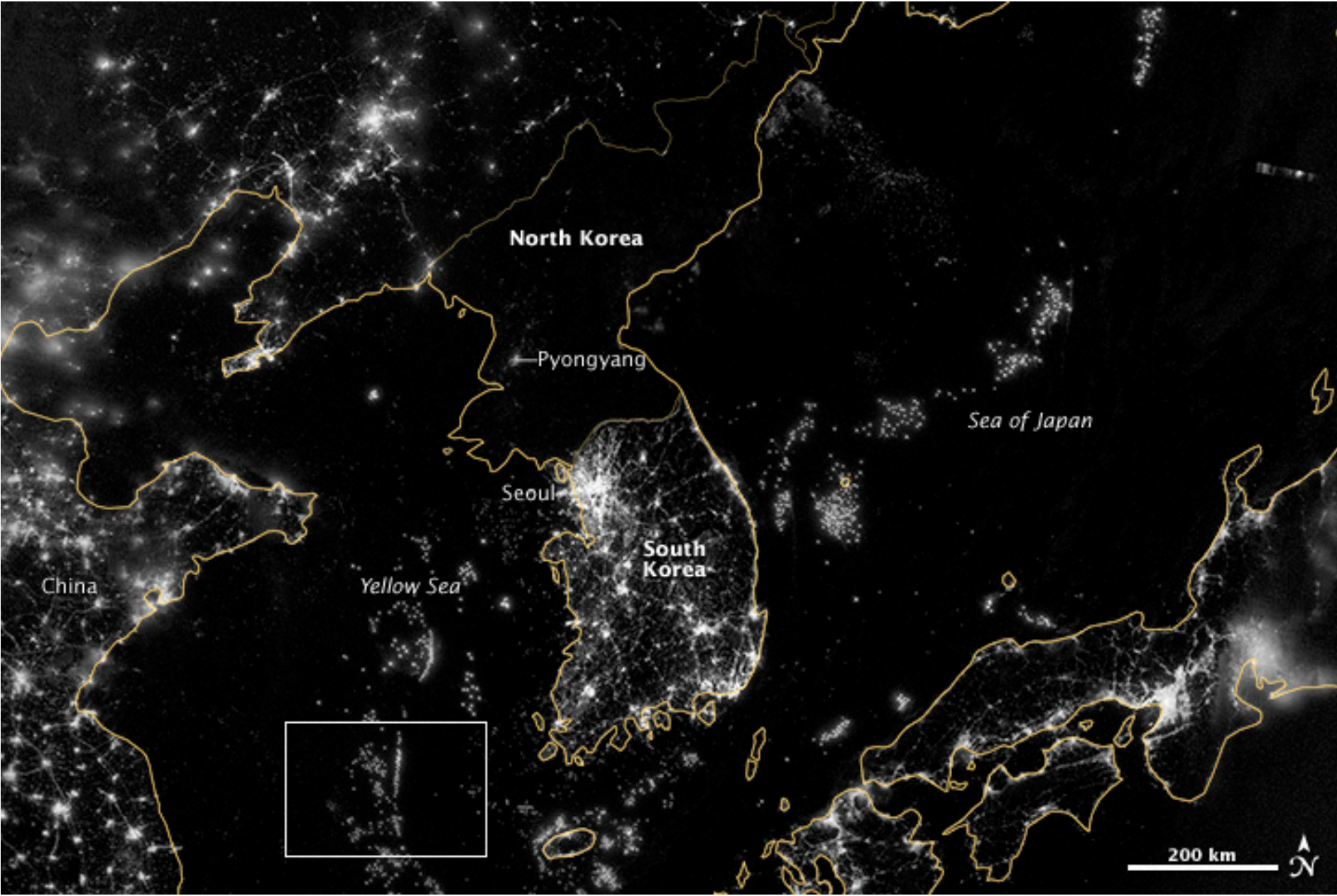
Table 1: **Taxonomy of Human-AI Collaboration Patterns.** We classify conversations into five distinct patterns across two broad categories based on how people integrate AI into their workflow.



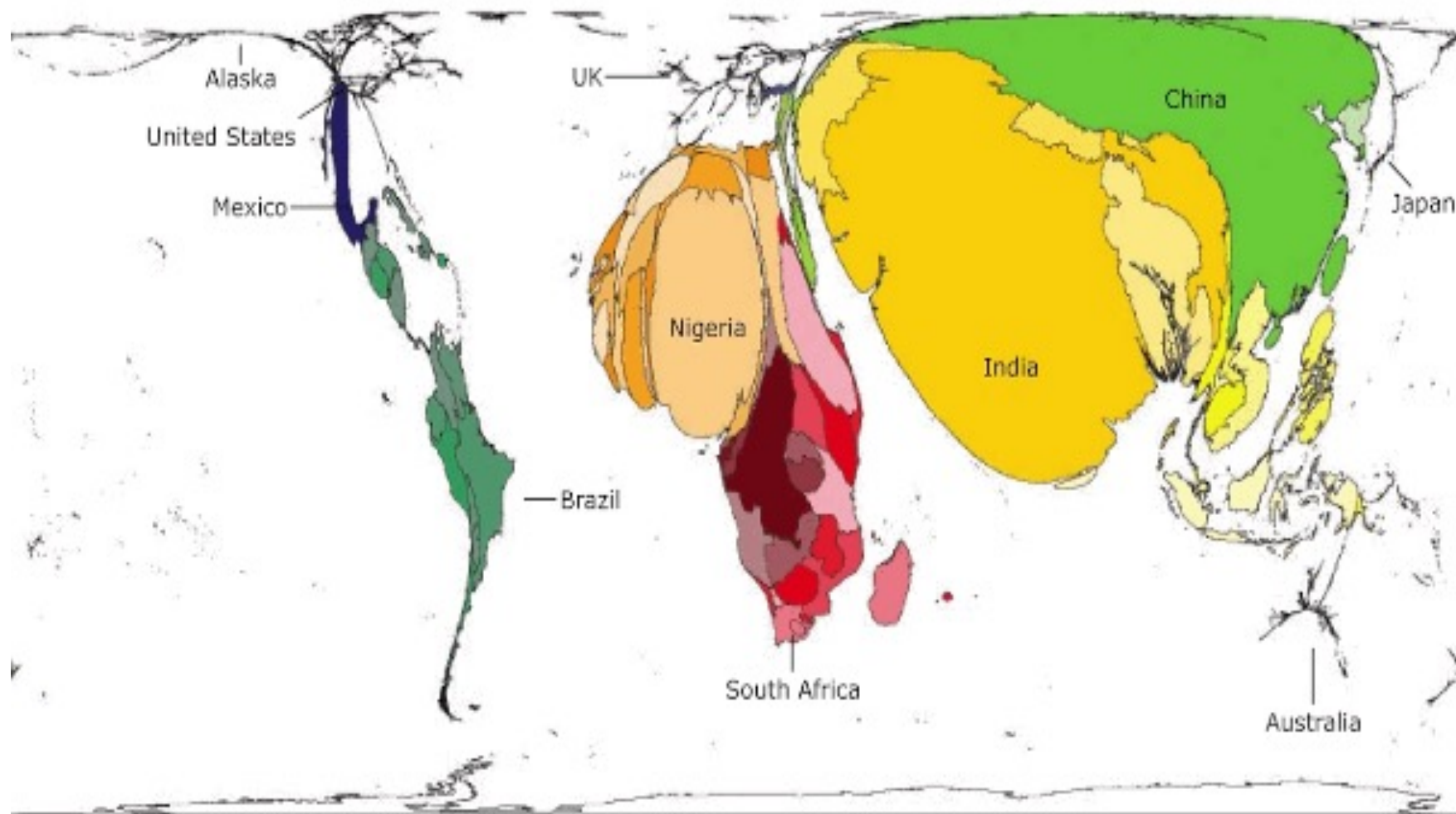
Gross Domestic Product (GDP, PPP) per Capita Around the World 2018







Worldmapper.org: An animation, from the world sized by people living on less than \$1 a day, to those earning over \$200



Less than \$1 a day

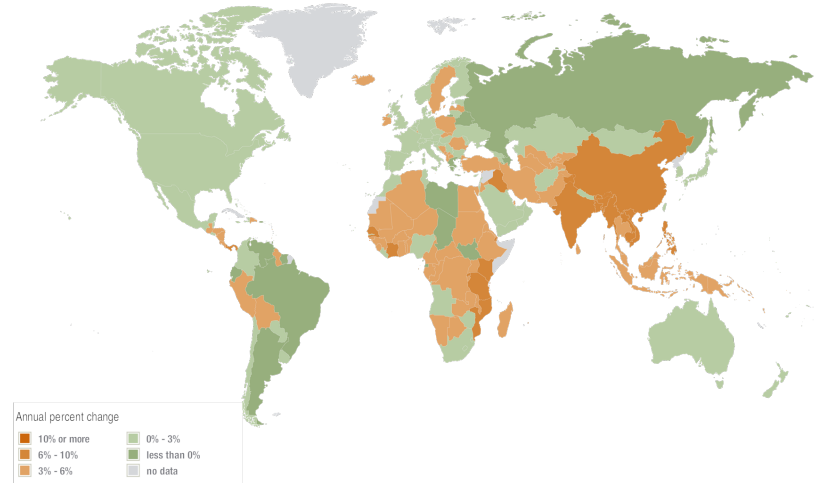
www.worldmapper.org

Principles and Concepts: Economic Development

- What is the real meaning of development?
- How can one apply economic concepts and theories to gain a better understanding of development process?
- Why do some countries develop and others remain poor?
- What are the sources of development and how do we measure development?

IMF Data Mapper ®

Real GDP growth (2016)
Source: World Economic Outlook (April 2016)



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Principles and Concepts: Economic Development

- Does historical record of development help us understand it better?
- What are the most influential theories of development and are they compatible?
- Is development process of developing nations independent or interdependent with that of developed nations?



Principles and Concepts: Economic Development

Examples of an affluent economy and subsistence economy.

Nature of development economics:

- Traditional economics - efficient allocation of scarce resources
- Political economics - social and political process
- Development economics - role of values, attitudes, and institutions

So, what do we mean by development?

Definition of Economic Development: 1950s

In economic terms, development is the capacity of a nation to generate and sustain an annual increase in its GNP of 5% or more.

Traditional economic measures:

- **GDP**: is the market value of all final goods and services produced within a country in a given period of time.
 - **$Y=C+I+G+NX$**
- **GNP**: is the market value of all final goods and services produced by permanent residents of a country in a given period of time.
 - **$GNP= GDP+ \textit{net factor income from abroad}$**

Definition of Economic Development:

- Common alternative index is the rate of growth of income per capita or per capita GNP.
 - **Per capita GNP**: is the per-head value of final goods and services produced by permanent residents of a country in a given period of time. It is converted to USD using the current exchange rate.
 - **PPP Measure**: the number of units of a country's currency required to purchase the same of basket of goods and services in the local market that a US \$1 would buy in the USA. Under PPP, exchange rates should adjust to equalize the price of a common basket of goods and services across countries. Penn World Tables rank countries using the PPP method.

The Big Mac index

Country	2000 — 2022	Under/over valued, %
Switzerland	Franc	30.3
Norway	Krone	21.6
Uruguay	Peso	18.1
Sweden	Krona	8.5
Canada	C\$	2.0
United States	US\$	BASE CURRENCY
Lebanon	Pound	-1.4
Israel	Shekel	-4.0
UAE	Dirham	-4.8
Euro area	Euro	-7.5
Australia	A\$	-10.2
Argentina	Peso	-11.3
Saudi Arabia	Riyal	-12.1
Britain	Pound	-13.8
New Zealand	NZ\$	-14.0
Brazil	Real	-17.5
Bahrain	Dinar	-17.6
Singapore	S\$	-17.7

Choose a base currency

US dollar

Show index at

Jun 2022

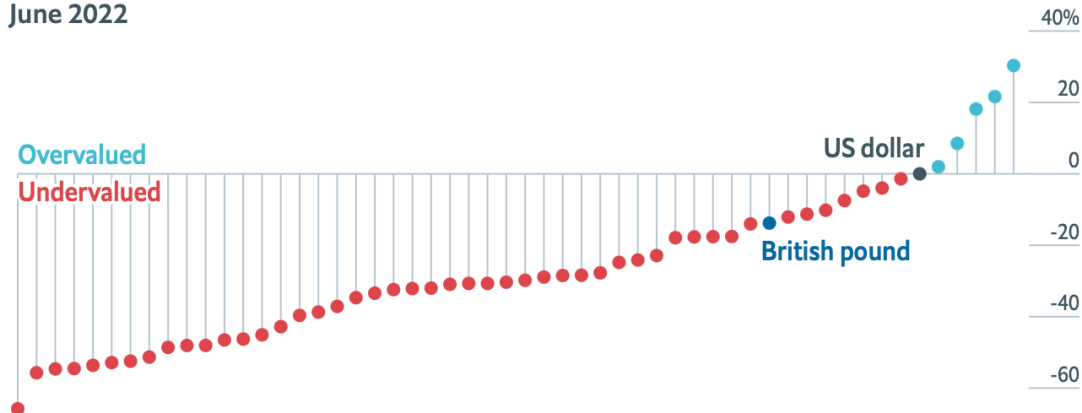
Adjust the index to account for GDP per person

Raw index

GDP-adjusted

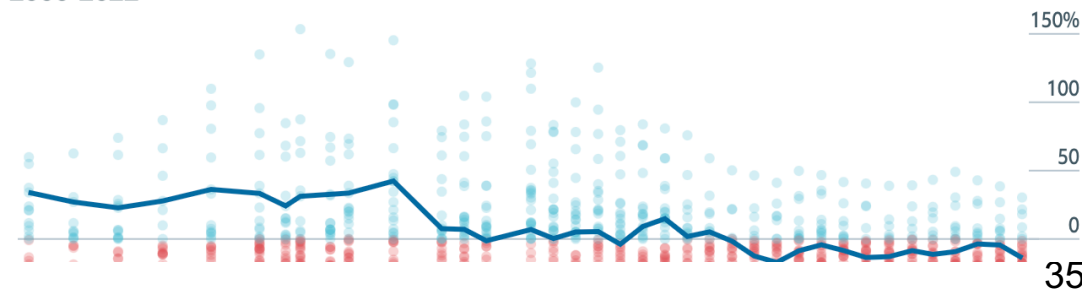
The British pound is 13.8% undervalued against the US dollar

June 2022



A Big Mac costs £3.69 in Britain and US\$5.15 in the United States. The implied exchange rate is 0.72. The difference between this and the actual exchange rate, 0.83, suggests the British pound is 13.8% undervalued

2000-2022



1. Basic concept: The index is based on the theory of PPP, which suggests that in the long run, exchange rates should adjust so that an identical basket of goods and services (in this case, a Big Mac) should cost the same in any two countries.
2. Calculation process: a) Find the price of a Big Mac in Country A (in local currency) b) Find the price of a Big Mac in the United States (in USD) c) Convert the local currency price to USD using the current exchange rate
3. Comparison:
 - If the converted price in Country A is higher than the US price, the currency is considered overvalued.
 - If the converted price in Country A is lower than the US price, the currency is considered undervalued.
4. Quantifying the over/undervaluation: The degree of over or undervaluation is calculated as the percentage difference between the converted local price and the US price.
5. Example: Let's say a Big Mac costs 4 USD in the US and 3.50 EUR in Germany. If the current exchange rate is 1 EUR = 1.2 USD: $3.50 \text{ EUR} * 1.2 \text{ USD/EUR} = 4.20 \text{ USD}$ The Euro would be considered overvalued by $(4.20 - 4.00) / 4.00 = 5\%$
6. Interpretation:
 - Overvalued: The local currency buys fewer Big Macs abroad than at home.
 - Undervalued: The local currency buys more Big Macs abroad than at home.

Comparison of GNP

	GNP Per Capita (US \$)	
Country	Exchange rate	PPP
UK	24,500	23,550
USA	34,260	34,260
Zimbabwe	480	2,590
Bangladesh	380	1,650
China	840	3,940
India	460	2,390
Sri Lanka	870	3,470

Classification of Economies

World Bank classifies 208 economies (population $\geq$ 30,000) based on gross national income per capita, 2020

- LIC $\leq$ \$1085
- LMC = \$1086-\$4255
- UMC = \$4256-\$13205
- High- income OECD $\geq$ \$13205
- Other high-income countries
- Different classification of other-high income countries

BY REGION	BY INCOME	BY LENDING
East Asia and Pacific	Low-income economies	IDA
Europe and Central Asia	Lower-middle-income economies	Blend
Latin America & the Caribbean	Upper-middle-income economies	IBRD
Middle East and North Africa	High-income economies	
North America		
South Asia		
Sub-Saharan Africa		

Economies are divided into IDA, IBRD, and Blend countries based on the operational policies of the World Bank. International Development Association (IDA) countries are those with low per capita incomes that lack the financial ability to borrow from the International Bank for Reconstruction and Development (IBRD). Blend countries are eligible for IDA loans but are also eligible for IBRD loans because they are financially creditworthy.

LOW-INCOME ECONOMIES (\$1,085 OR LESS)

[28]

Afghanistan	Guinea-Bissau	Somalia
Burkina Faso	Korea, Dem. People's Rep	South Sudan
Burundi	Liberia	Sudan
Central African Republic	Madagascar	Syrian Arab Republic
Chad	Malawi	Togo
Congo, Dem. Rep	Mali	Uganda
Eritrea	Mozambique	Yemen, Rep.
Ethiopia	Niger	Zambia
Gambia, The	Rwanda	
Guinea	Sierra Leone	

Classification of Economies

<u>East Asia and the Pacific:</u> Korea, Rep China Indonesia	UMC LMC LIC
<u>Europe and Central Asia:</u> Czech Republic Lithuania Armenia	UMC LMC LIC
<u>Latin America and the Caribbean:</u> Brazil Cuba Nicaragua	UMC LMC LIC

Classification of Economies

<u>Middle East and North Africa:</u> Bahrain Iraq Yemen, Rep	UMC LMC LIC
<u>South Asia:</u> Sri Lanka India	LMC LIC
<u>Sub-Saharan Africa:</u> South Africa Namibia Ethiopia	UMC LMC LIC

Classification of Economies

High Income OECD:

Australia

Canada

USA

Japan

Denmark

France

Switzerland

UK

Other high income countries:

The Bahamas

Kuwait

Singapore

Problems:

- Underreporting of income by developing countries.
- High proportion of income is generated for self-consumption.
- Prices of non-traded goods are not appropriately reflected in exchange rates.
- Markets are not competitive and externalities are not reflected.

Definition of Economic Development: 1970s



Dethronement of GNP in the 1970s and increasing emphasis on “redistribution from growth”.



Increasing emphasis on non-economic social indicators.



Economic development consists of the reduction or elimination of **poverty**, **inequality** and **unemployment** within the context of a growing economy.

Human goals of economic development :

Sen's "Capabilities" Approach: 1985

- Economic growth is not an end in itself and has to enhance the lives people lead and the freedoms that they enjoy.
- The concept of "capability to function" is integral to the Capability Approach, which was primarily developed by the Nobel laureate economist Amartya Sen and philosopher Martha Nussbaum.
 - This framework views people's capabilities – their real freedoms to do and be what they value – as the ultimate units of human well-being.

Human goals of economic development :

Sen's "Capabilities" Approach: 1985

- **Capability to function** is what matters for status as a poor/non-poor person and it goes beyond availability of commodities.
 - **Capabilities:** “freedom that a person has in terms of the choice of his functionings...”
 - For example, consider two individuals – one who can't read because they never had access to education, and another who can read but chooses not to. Even though both individuals are not currently "functioning" as readers, their capabilities are different. The first person is non-literate because they do not have the capability to read due to lack of access to education, while the second person is voluntarily non-literate – they have the capability to read, but choose not to. In terms of capability, the second person is in a better situation, as they at least have the option to read.
 - **Functioning** is what a person does with commodities of given characteristics that they possess/control
 - It refers to the various things a person may value doing or being, like being in good health, being educated, being safe, participating in community life, etc. A functioning represents parts of the state of a person – in particular, the various things they can do or be in leading a life.

- “Capability to function”, highlights the importance of not just the availability of commodities, but also the freedom and ability to use these commodities in ways that genuinely improve one's life.
- The key idea is that economic growth and development should not just focus on the accumulation of wealth, but on expanding people's capabilities – their freedom to lead the kind of life they value, and have reason to value. Economic growth is merely a means to that end, and not an end in itself.
- Amartya Sen traced five sources of disparity between real incomes and actual advantages:
 - Personal heterogeneities
 - Environmental diversities
 - Social climate variations
 - Differences in relational perspectives
 - Distribution within family

- Amartya Sen traced five sources of disparity between real incomes and actual advantages:
 - Personal heterogeneities: People have different physical and mental characteristics, which can affect their ability to convert income into quality of life.
 - Environmental diversities: The environment in which a person lives can greatly affect how well they can convert income into quality of life.
 - Social climate variations: The society in which a person lives can affect their quality of life, independent of their income. This includes factors like public health and education systems, crime levels, social networks and community support, cultural norms, etc.
 - Differences in relational perspectives: This refers to how individuals see themselves in relation to others in society.
 - Distribution within family: This refers to how resources are allocated within a household or a family.

Definition of Economic Development: 1990s



World Bank in its 1991 WDR asserted that the “challenge of development is to improve the ***quality of life*** (QoL).”



The improved QoL involves higher incomes, better education, higher standards of health and nutrition, less poverty, a cleaner environment, more equality of opportunities, greater individual freedom, and a richer cultural life.

Economic factors

- Capital
- Labor
- Natural resources
- Technology
- Established markets (labor, financial, goods)

Non-economic factors (institutional, social, values)

- attitudes toward life and work
- public and private structures
- cultural traditions
- systems of land tenure, property rights
- integrity of government agencies

Definition of Economic Development:

“Development is a multi dimensional process involving changes in social structures, popular attitudes, and national institutions, as well as the acceleration of economic growth, the reduction of inequality, and the eradication of poverty.” (Todaro and Smith)



Development is both a physical reality and a state of mind for attaining a better life.



Three basic core values as a practical guideline for understanding development

Sustenance
Self-esteem
Freedom



Specific components of better life vary from time to time and from society to society.



Three Objectives of Development:

Increase availability and distribution of basic goods
Raise levels of living
Expand range of social and economic choices available to individuals

Human Development Index

Initiated in 1990 and undertaken by United Nations Development Programme (UNDP) in its annual series of Human Development Report (HDR).



HDI is based on 3 goals:

Longevity

Knowledge

Standard
of living



$$\text{HDI} = \frac{1}{3}(\text{Income index}) + \frac{1}{3}(\text{Life expectancy index}) + \frac{1}{3}(\text{education index})$$

Human Development Index

- Ranks 175 countries into 3 groups
 - Low human development = 0.00-0.099
 - Medium human development = 0.5-0.799
 - High human development = 0.80-1.00

Country	HDI	GDP rank-HDI rank
Low HD: Tanzania	0.436	+21
Medium HD: Turkey	0.735	-21
High HD: Canada	0.936	+3

Structural Diversity of Developing Economies

Size (geographical, population) and income of the country

Historical and colonial background

Physical and HR endowments

Ethnic and religious composition

Relative importance of public and private sectors

Nature of its industrial structure

Degree of dependence on external economic and political forces

Distribution of power, institutional and political structures

Structural Diversity of Developing Economies:

Important observations

- There is no necessary relation among a country's size, per capita income, and degree of equality/ inequality
- Colonialism exposed African and Asian countries to new forms of potential exploitation and have resulted in different institutional and social patterns
- Except for the Persian Gulf oil states, most developing countries have minimal endowments of raw materials and minerals

Structural Diversity of Developing Economies:

Important observations

- Ethnic and religious conflicts have become frequent post cold war. Indigenous populations continue to be discriminated against in Latin American countries. Malaysia and Mauritius are exceptions.
- Most developing economies have mixed economic systems and the degree of corruption differs widely across countries.
- Majority of developing countries are agrarian and agriculture is a way of life for the people.

Common Characteristics of Developing Nations:

- Low levels of living and income inequality
- Extent of Poverty
- Health characterized by high IMRs and malnutrition
- Low levels of education and literacy
- Low levels of labor productivity
- High rates of population growth and dependency burdens
- Substantial dependence on agricultural production and primary product exports
- Prevalence of imperfect markets and incomplete information
- Dominance, dependence, and vulnerability in international relations

Distribution of Income

Table 2.6 Global Income Disparity between the Richest and Poorest 20 Percent of the World's Population, 1960–2000

Year	Ratio of Income Shares
	Richest to Poorest
1960	30 to 1
1970	32 to 1
1980	45 to 1
1991	61 to 1
2000	70 to 1

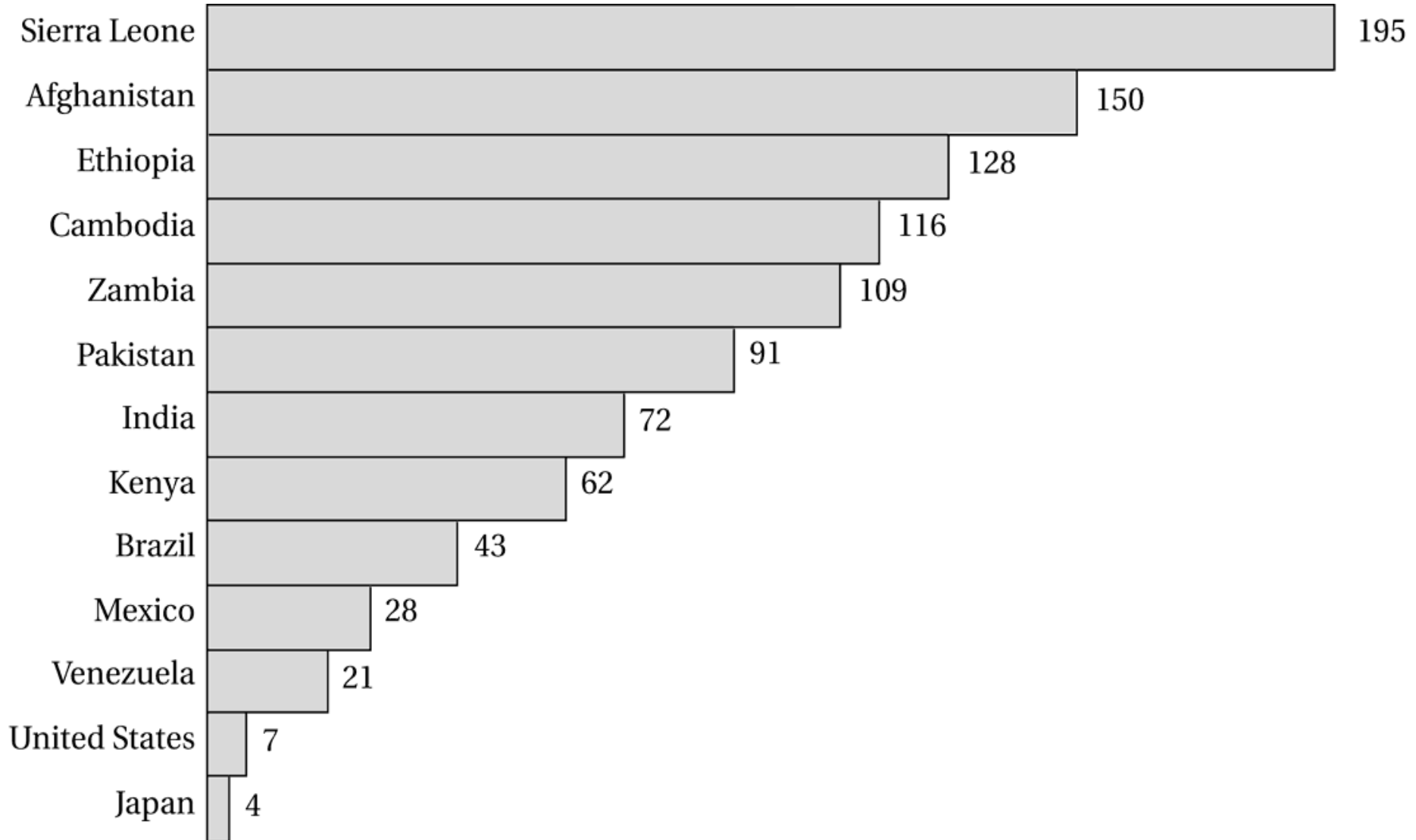
Sources: United Nations Development Program, *Human Development Report, 1992* (New York: Oxford University Press, 1992), p. 36; United Nations Development Program, *Human Development Report, 1994 and 2001* (New York: Oxford University Press, 1994 and 2001).

Poverty: Share of population living on < \$1/ day

Region	1990	1998
East Asia and Pacific	27.6	15.3
Europe and Central Asia	1.6	5.1
Latin America and the Caribbean*	16.8	15.6
Middle East and N. Africa	2.4	1.9
South Asia*	44	40
Sub-Saharan Africa*	47.7	46.3
Total	29	24

* Increase in absolute number of people

Health: Infant Mortality Rates in Selected Countries, 1998 (per 1,000 live births)



Health and Education: Indicators

Table 2.8 Human Health and Education Deprivation in the Developing World, 2001

Nature of Health Deprivation	People Deprived
Lack of access to health services	766 million (1995)
Lack of access to safe water	968 million (1998)
Lack of access to sanitation	2.4 billion (1998)
Children dying before age 5 from preventable causes	11 million (1998)
Underweight children under age 5	163 million (1998)
People living with HIV/AIDS	34 million (2000)
Illiterate adults	854 million (2000)
Children not in school	325 million (2000)

Source: United Nations Development Program, *Human Development Report, 2001* (New York: Oxford University Press, 2001), p. 9. Reprinted with permission.

HDI and GDP Ranks

TABLE 2.9 Human Development Index for Twenty-Two Selected Countries, 1999

Country	Relative Ranking (lowest to highest)	Human Development Index (HDI)	Real 1995 GDP Per Capita (PPP\$)	GDP Rank minus HDI Rank ^a
Low human development				
Sierra Leone	162	0.258	448	0
Ethiopia	157	0.321	628	0
Malawi	151	0.397	586	+8
Guinea	150	0.397	1,934	-32
Angola	146	0.422	3,179	-44
Tanzania	140	0.436	501	+21
Bangladesh	132	0.470	1,483	-4
Medium human development				
India	115	0.571	2,248	0
Nicaragua	106	0.634	2,279	+7
China	87	0.718	3,617	+7
South Africa	94	0.702	8,908	-49
Peru	73	0.743	4,622	+8
Oman	71	0.747	13,356	-33
Turkey	82	0.735	6,380	-21
Malaysia	56	0.774	8,209	-4
Thailand	66	0.757	6,132	-3
High human development				
Kuwait	43	0.818	17,289	-14
Costa Rica	41	0.821	8,860	+6
United Kingdom	14	0.923	22,093	+5
United States	6	0.934	31,872	-4
Canada	3	0.936	26,251	+3
Norway	1	0.939	28,433	+2

Source: United Nations Development Program, *Human Development Report, 2001* (New York: Oxford University Press, 2001), annex tab. 1. Reprinted with permission.

^aA positive figure indicates that the HDI rank is better than the real GDP per capita (PPP\$) rank; a negative indicates the opposite.

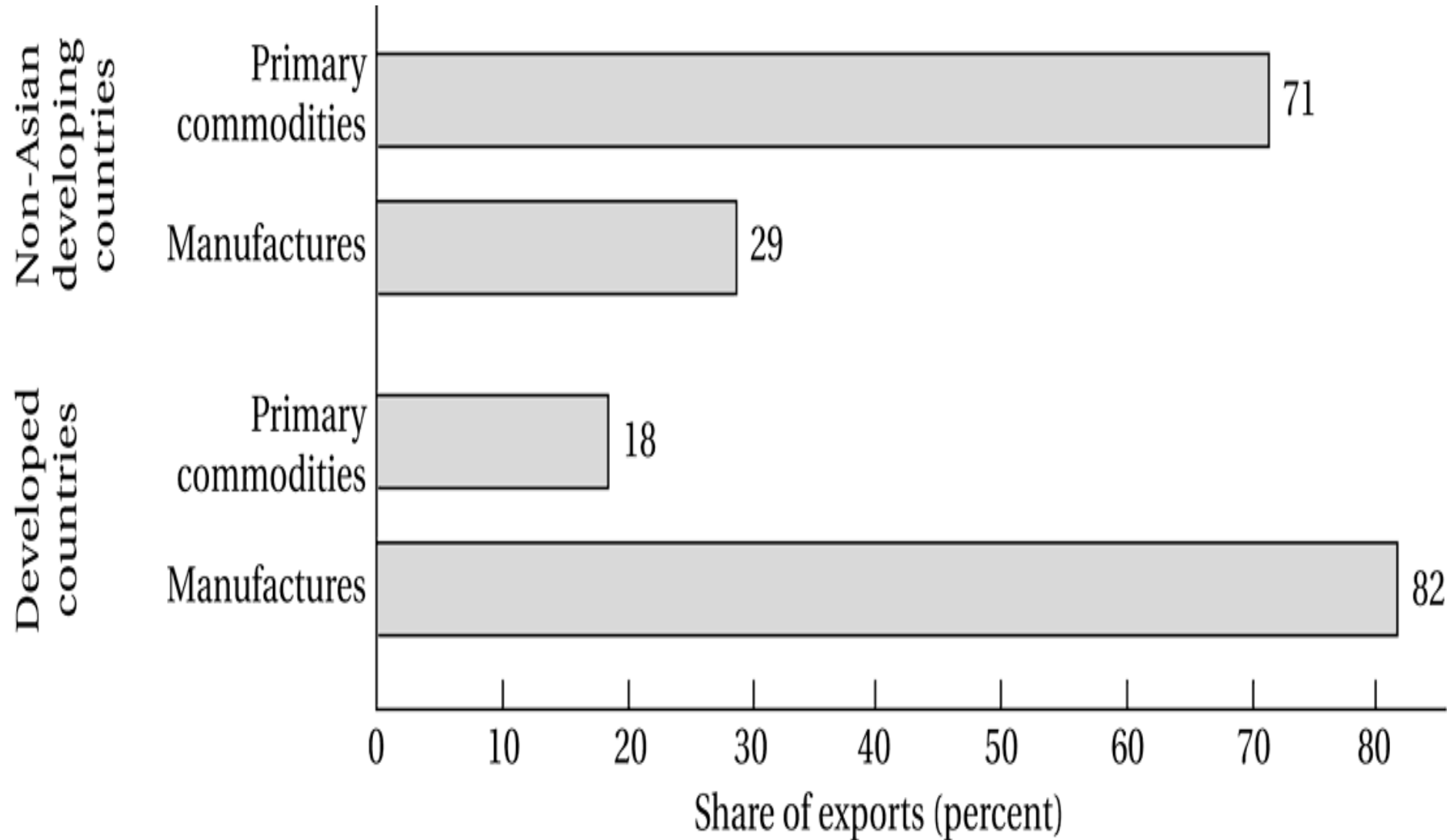
Population and Labor Force

TABLE 2.12 Population, Labor Force, and Production in Developed and Less Developed Regions, 1995–1997

Region	Population (millions)	Urban (%)	Rural (%)	Labor Force in Agriculture (%)	Agricultural Share of GNP (%)
World	5,840	43	57	49	—
Developed					
countries	1,175	74	26	5	3
Europe	729	72	28	7	5
North America	298	75	25	3	2
Japan	126	78	22	7	2
Less developed					
countries	4,666	36	64	58	14
Africa	743	31	69	68	20
South Asia	1,417	27	73	64	30
East Asia	1,958	35	65	70	18
Latin America	490	72	28	25	10

Sources: Population Reference Bureau, 1997 *World Population Data Sheet* (Washington, D.C.: Population Reference Bureau, 1997); *World Bank, World Development Report, 1997* (New York: Oxford University Press, 1997), tabs. 4 and 12.

Composition of World Exports (percentages of primary and manufactured products)



Composition of World Exports (percentages of primary and manufactured products)

Markets are imperfect in developing countries due to:

- Economies of scale
- Thin markets for many products due to limited demand
- Widespread externalities (good and bad)
- Prevalence of common property resources
- Incomplete information
- Information is costly to obtain

Case Study: Brazil

- Brazil is the largest and most populous country in Latin America, with close to 200 million people.
- Brazil's growth from the 1960s through 1980s was the best in Latin America. [link](#)
- "Brazil is a country of the future – and always will be", Brazil has experienced "growth without development". [link](#)
- In Brazil:
 - World-competitive industry coexists with stagnant, protected sectors;
 - Modern agriculture coexists with low-productivity traditional practices;
 - A higher percentage of poverty than would be expected for an upper-middle-income country; [link](#)
 - One of the countries with the highest levels of inequality in the world;



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